APPSC GROUP 1 / GROUP 2 — GEOGRAPHY

Volume 2: Physical Geography of India and Andhra Pradesh

Location · Physiography (Himalaya, Plains, Plateau, Desert, Coasts, Islands) · Drainage · Climate and Monsoon · Soils · Natural Vegetation and Wildlife · Natural Hazards and Disaster Management

READ THIS FIRST

How to use: same pattern as Volume 1: **Blue** = simple explanation · **Orange** = Andhra Pradesh / Kurnool link · **Red** = exam traps · **Green** = 2-minute revision · **Purple** = self-check. Use the **maps** while reading and later try to redraw them from memory.
Study plan (6 days): Day 1 Ch 1-2 · Day 2 Ch 3 (Himalayas and plains) · Day 3 Ch 4 (rivers) · Day 4 Ch 5 (climate) · Day 5 Ch 6-7 (soil, vegetation) · Day 6 Ch 8 (hazards) + final test.
Syllabus match (Group 1 and 2): 'India and Andhra Pradesh: Major physiographic features, Climate, Drainage System, Soils and Vegetation, Natural Hazards and Disasters and their Management'. This volume covers all of it.
Note: maps are schematic (not to exact scale). Figures marked (verify) can change; recheck them in the latest current affairs notes.

CHAPTER 1 — INDIA: LOCATION, SIZE AND NEIGHBOURS

Where is India on the globe?

IN SIMPLE WORDS (what / how / why)

Latitude: 8°4′N (Kanyakumari, mainland) to 37°6′N. **Longitude:** 68°7′E to 97°25′E. The **Tropic of Cancer (23.5°N)** divides India almost into two equal halves: the south is **tropical**, the north is **sub-tropical**.
Size: about **3.28 million sq km**, the **7th largest country** in area (Russia, Canada, China, USA, Brazil, Australia are bigger), about **2.4%** of the world's land but about **17.5%** of its people.
Why is India's location important? (1) Central position in Asia between the East and West. (2) The **Indian Ocean** gives sea routes. (3) The **Himalaya** keeps cold Siberian winds out and gives snow-fed rivers. (4) The **Deccan** juts into the ocean, so the Indian Ocean is named after India.
Southernmost point of India = Indira Point (Great Nicobar) at about 6°45′N (mainland: Kanyakumari). **Standard time meridian = 82°30′E** (near Mirzapur, UP).

Neighbours (7 land + 2 sea)

Type	Countries	Remember
Land (7)	Pakistan, Afghanistan, China, Nepal, Bhutan, Myanmar, Bangladesh	**Longest border: Bangladesh** (about 4,100 km). Afghanistan border is the shortest (the Wakhan side) and is only through PoK as per India's position.
Sea (2)	Sri Lanka, Maldives	**Palk Strait** (India–Sri Lanka); **Gulf of Mannar**; **Ten Degree Channel** (Andaman | Nicobar); **Eight Degree Channel** (Minicoy | Maldives).
Border lines	Radcliffe (India–Pakistan, 1947); Durand (Afghanistan–Pakistan); McMahon (India–China, east)	LAC = Line of Actual Control with China.

Coastline: about **7,516 km** including islands (mainland about 6,100 km). **9 coastal states:** Gujarat (longest), Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, **Andhra Pradesh**, Odisha, West Bengal. Coastal UTs: Puducherry, Lakshadweep, Andaman & Nicobar, Dadra & Nagar Haveli and Daman & Diu.
Land boundary: about 15,100 km, shared by 17 states/UTs (figures verify).

ANDHRA PRADESH / KURNOOL CONNECTION

Andhra Pradesh coast: about 1,053 km (among the longest of Indian states; Gujarat is first; verify rank). AP lies between about **12°41′N and 19°55′N** and **76°45′E and 84°46′E** (approx.), entirely in the **tropical zone, south of the Tropic of Cancer**.
Kurnool is at about **15.8°N, 78°E**: inland, on the **Deccan plateau**, in the **rain shadow** zone.

2-MINUTE REVISION

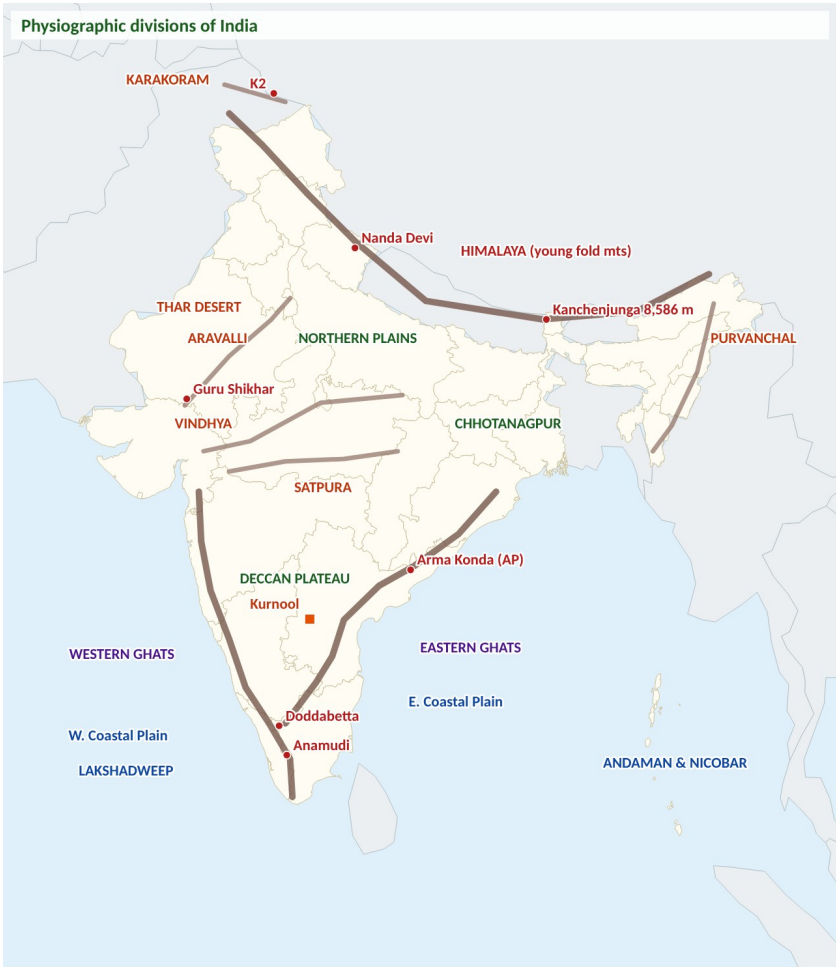
Lat 8°4′N-37°6′N · Long 68°7′E-97°25′E · Tropic of Cancer halves India · 7th largest · 7 land neighbours · Indira Point south · IST 82.5°E · coastline 7,516 km · 9 coastal states · AP coast about 1,053 km.

CHECK YOURSELF

Q1. Longest land border with: **Bangladesh** · Q2. Channel between Minicoy and Maldives: **Eight Degree Channel** · Q3. State with longest coastline: **Gujarat**

CHAPTER 2 — PHYSIOGRAPHIC DIVISIONS OF INDIA

Six big landform regions: how and why they formed



Map 2.1 Physiographic divisions (schematic)

IN SIMPLE WORDS (what / how / why)

Physiography = study of the natural features of land (mountains, plains, plateaus). India has **six** divisions: **(1) Northern Mountains (Himalaya), (2) Northern Plains, (3) Peninsular Plateau, (4) Indian (Thar) Desert, (5) Coastal Plains, (6) Islands.** **Why so different?** Remember Volume 1: India was part of **Gondwanaland** and is the **oldest, hardest** block (the plateau). It then drifted north and **collided** with Asia, **pushing up the Himalaya** (young). The rivers from the Himalaya then **filled the gap** in between with silt and made the plains. So: **old plateau + young mountains + plains made by rivers.**

2.1 The Himalaya (Northern Mountains)

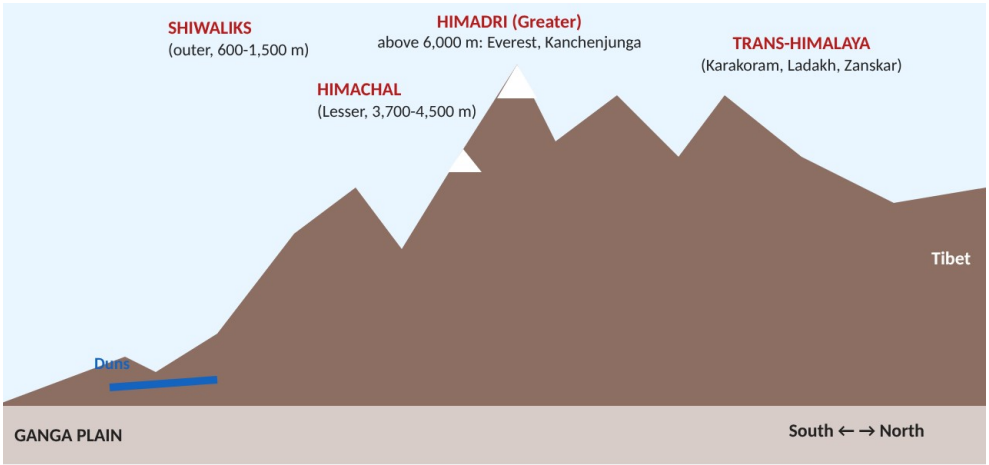


Fig 2.2 Cross-section of the Himalaya (south to north)

IN SIMPLE WORDS (what / how / why)

Young fold mountains, about **2,400 km long**; width 400 km (Kashmir) to 150 km (Arunachal). They are still rising, so earthquakes

and landslides are common. They are in **three parallel ranges** (south to north):
1. Shiwaliks (Outer Himalaya) — lowest, 600-1,500 m; made of loose river sediments; **Duns** (valleys, e.g. Dehradun) between them and the Lesser Himalaya. **2. Himachal / Lesser Himalaya** — 3,700-4,500 m; **Pir Panjal** (longest range), Dhauladhar, Mussoorie; **valleys of Kashmir, Kullu, Kangra**; **hill stations** Shimla, Mussoorie, Nainital, Darjeeling. **3. Himadri / Greater Himalaya** — highest, average above 6,000 m, always snow-covered: **Kanchenjunga (8,586 m; India's highest peak, 3rd in the world)**, **Nanda Devi (7,816 m; highest peak entirely in India)**; Mt Everest is in Nepal. Beyond it is the **Trans-Himalaya**: **Karakoram (K2 = Godwin-Austen, 8,611 m, world's 2nd highest; in PoK/Gilgit, claimed by India)**, Ladakh and Zanskar ranges.

Pass	Connects	Note
Zoji La	Srinagar – Leh	Kashmir to Ladakh
Banihal	Jammu – Srinagar	Jawahar tunnel
Khardung La	Leh – Nubra	One of the highest motorable roads
Rohtang (Atal Tunnel)	Manali – Lahaul	Atal tunnel (2020) under it
Shipki La	Himachal – Tibet	Sutlej enters India
Nathu La, Jelep La	Sikkim – Tibet	Border trade with China
Lipulekh	Uttarakhand – Tibet	Kailash Mansarovar route
Bomdi La	Arunachal – Tibet	

Glaciers: Siachen (longest outside polar regions, Karakoram), **Gangotri** (source of the Ganga's Bhagirathi), **Yamunotri**, **Zemu** (Sikkim).
Eastern hills (Purvanchal): Patkai, Naga, Manipur, Mizo (Lushai) hills, and **Garo-Khasi-Jaintia** (Meghalaya plateau, the wettest region). Rivers: Barak.
Regional divisions by rivers: Punjab Himalaya (Indus–Sutlej), Kumaon (Sutlej–Kali), Nepal (Kali–Tista), Assam (Tista–Dihang).

2.2 The Northern Plains

IN SIMPLE WORDS (what / how / why)

What: a vast flat land of **alluvium** (silt, sand, clay) about **2,400 km long and 240-320 km wide**, formed by the **Indus, Ganga and Brahmaputra** carrying silt from the Himalaya for millions of years (the sea-like trough was slowly filled).
Four belts, from the mountain foot to the south: Bhabar (pebbles at the foothill, streams disappear underground) → **Terai** (the water comes out again; swampy, forests, now farms) → **Bhangar** (older alluvium, higher, has **kankar** lime nodules) → **Khadar** (newer, flood-fed alluvium, very fertile).
Parts: Punjab plains (west, five rivers), **Ganga plains** (largest; Haryana to Bengal), **Brahmaputra plains** (Assam). **Doab** = land between two rivers.
Importance: most fertile region, densest population; wheat, rice, sugarcane, jute. Often flooded.

2.3 The Peninsular Plateau

IN SIMPLE WORDS (what / how / why)

What: a **triangular, very old, hard** block of crystalline rock (part of Gondwana). Its edges are the **Aravalli** (NW), **Vindhya and Satpura** (north), **Western Ghats** (west) and **Eastern Ghats** (east).
Two parts: Central Highlands (north of Narmada: **Malwa**, Bundelkhand, Baghelkhand, **Chhotanagpur** plateau rich in minerals) and **Deccan Plateau** (south of Narmada; tilted **west to east**, so most big rivers flow to the **Bay of Bengal**).
Deccan Traps: about 65 million years ago huge **basalt lava** flows covered much of the plateau; weathering made the **black cotton soil (regur)**.
Aravalli = oldest fold mountains of India (worn down); highest peak **Guru Shikhar (1,722 m, Mt Abu)**. Vindhya and Satpura are block/residual hills with the **Narmada–Tapti rift valleys** between them.

	Western Ghats (Sahyadri)	Eastern Ghats
Shape	Continuous, a steep wall facing the Arabian Sea	Broken, discontinuous; cut by the Godavari, Krishna, Mahanadi and Kaveri
Height	Higher: **Anamudi 2,695 m** (Anaimalai) — highest peak of South India; Doddabetta 2,637 m (Nilgiris)	Lower: **Jindhagada / Arma Konda about 1,680 m (AP)**; Mahendragiri (Odisha) 1,501 m
Rain	Heavy on the west (windward)	Gets less SW monsoon; gets NE monsoon rain
Gaps	**Thal, Bhor, Pal Ghat** passes	Few passes, many river gaps
Meet	They meet in the **Nilgiri hills**	
Other	UNESCO World Heritage Site, biodiversity hotspot	Nallamala, Seshachalam, Velikonda, Palkonda, Javadi, Shevaroy